

Area Under The Curves

Question1

The area of the region (in sq units) bounded by the curves $x^2 + y^2 = 16$ and $y^2 = 6x$ is

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

$$4\pi + 4\sqrt{3}$$

B.

$$\frac{2}{3}(4\pi + \sqrt{3})$$

C.

$$\frac{4}{3}(4\pi + \sqrt{3})$$

D.

$$\frac{4\pi + \sqrt{3}}{3}$$

Answer: C

Solution:

Given:

The circle is $x^2 + y^2 = 16$.

The parabola is $y^2 = 6x$.

Finding Intersection Points:

To find where these curves meet, put y^2 from the parabola into the circle's equation:

$$\begin{aligned}
 \rightarrow x^2 + y^2 &= 16 \\
 x^2 + 6x &= 16 \\
 x^2 + 6x - 16 &= 0 \\
 (x + 8)(x - 2) &= 0 \\
 x &= -8, 2
 \end{aligned}$$

We only use $x = 2$ (since $x = -8$ is outside the region for $y^2 = 6x$ and gives no real y for $x^2 + y^2 = 16$).

For $x = 2$, $y^2 = 6 \cdot 2 = 12$. So $y = \sqrt{12}$ and $y = -\sqrt{12}$.

Finding the Area:

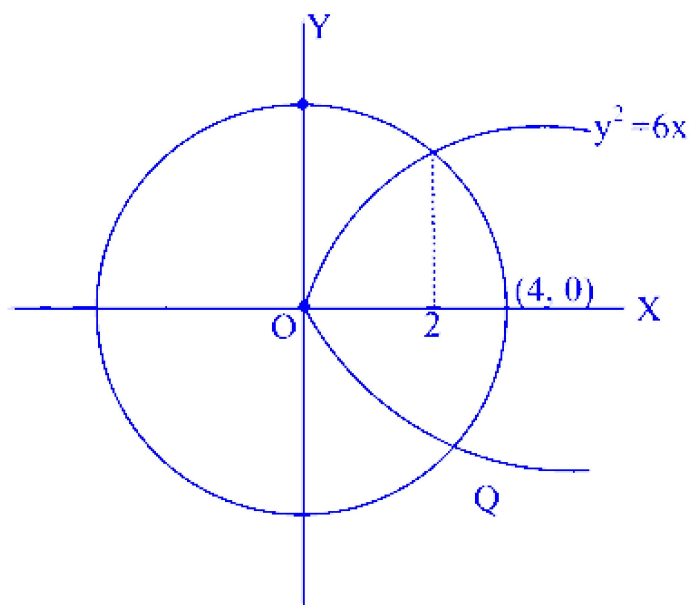
We want the area inside the circle and outside the parabola, between $x = 0$ and $x = 4$.

To get this area, we split it into two parts:

- From $x = 0$ to $x = 2$ (where the parabola is below the circle), the top curve is the parabola: $y = \sqrt{6x}$
- From $x = 2$ to $x = 4$ (where only the circle remains), the top curve is the circle: $y = \sqrt{16 - x^2}$

So, the area is:

$$\text{Area} = 2 \left[\int_0^2 y_{\text{parabola}} dx + \int_2^4 y_{\text{circle}} dx \right]$$



$$\text{So, } = 2 \int_0^2 \sqrt{6x} dx + 2 \int_2^4 \sqrt{16 - x^2} dx$$

Solving the Integrals:

First, $\int_0^2 \sqrt{6x} dx$:

$\sqrt{6x} = \sqrt{6}\sqrt{x}$, so

$$\int_0^2 \sqrt{6x} dx = \sqrt{6} \int_0^2 x^{1/2} dx = \sqrt{6} \cdot \left[\frac{2}{3} x^{3/2} \right]_0^2 = \sqrt{6} \cdot \frac{2}{3} [(2)^{3/2} - 0]$$

Second, $\int_2^4 \sqrt{16 - x^2} dx$:

The formula for this is:

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right)$$

Here $a = 4$. Plug in $x = 4$ and $x = 2$ to get the answer.

Now, combine both solutions:

$$\begin{aligned} &= 2\sqrt{6} \cdot \frac{2}{3} \left(2^{3/2} \right) + 2 \left[\frac{x}{2} \sqrt{16 - x^2} + 8 \sin^{-1} \left(\frac{x}{4} \right) \right]_{x=2}^{x=4} \\ &= \frac{4}{3} \sqrt{6} \cdot 2\sqrt{2} + 2 \left[\left(0 + 8 \frac{\pi}{2} \right) - \left(2\sqrt{3} + 8 \frac{\pi}{6} \right) \right] \\ &= \frac{16\sqrt{3}}{3} + 2 \left(\frac{8\pi}{3} - 2\sqrt{3} \right) \\ &= \frac{16\sqrt{3}}{3} + \frac{16\pi}{3} - 4\sqrt{3} \\ &= \frac{4\sqrt{3}}{3} + \frac{16\pi}{3} \\ &= \frac{4}{3} (4\pi + \sqrt{3}) \end{aligned}$$

Question2

The area (in sq. units) of the region bounded by the curves $y = x^2$ and $y = 8 - x^2$ is

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

$$\frac{32}{3}$$

B.

$$\frac{16}{3}$$

C.

$$\frac{64}{3}$$

D.

$$\frac{128}{3}$$

Answer: C

Solution:

Given, curves are $y = x^2$ and

$$y = 8 - x^2$$

Equating these two equations, we get

$$x^2 = 8 - x^2 \Rightarrow 2x^2 = 8$$

$$\Rightarrow x^2 = 4 \Rightarrow x = \pm 2$$

$$\begin{aligned}\text{Thus, area} &= \int_{-2}^2 [(8 - x^2) - x^2] dx \\ &= \int_{-2}^2 (8 - x^2 - x^2) dx = \int_{-2}^2 (8 - 2x^2) dx \\ &= \left[8x - \frac{2}{3}x^3 \right]_{-2}^2 \\ &= \left\{ 8(2) - \frac{2}{3}(2)^3 \right\} - \left\{ 8(-2) - \frac{2}{3}(-2)^3 \right\} \\ &= \left(16 - \frac{16}{3} \right) - \left(-16 + \frac{16}{3} \right) \\ &= 16 - \frac{16}{3} + 16 - \frac{16}{3} \\ &= 32 - \frac{32}{3} = \frac{96-32}{3} = \frac{64}{3}\end{aligned}$$

$\therefore$ The area of the region bounded by the curves is $\frac{64}{3}$ sq. units.

Question3

Area of the region (in sq. units) bounded by the curve $y = x^2 - 5x + 4$, $x = 0$, $x = 2$ and the X -axis is

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

$$\frac{8}{3}$$

B.

3

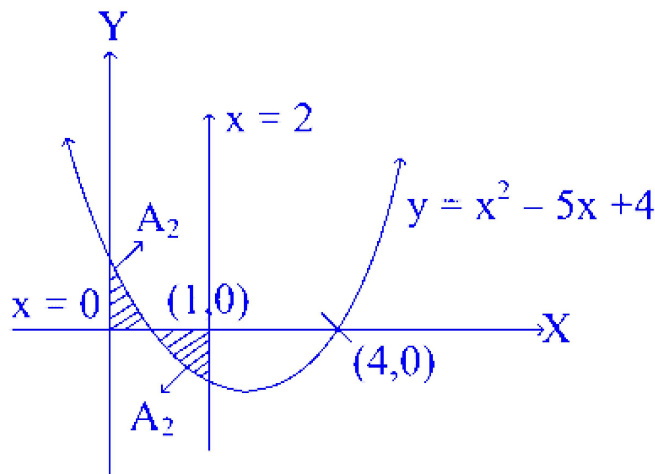
C.

D.

$$\frac{5}{2}$$

Answer: B**Solution:**

$$y = x^2 - 5x + 4 = (x - 1)(x - 4)$$



$$\begin{aligned}
 A &= A_1 + A_2 = \int_0^1 (x^2 - 5x + 4) dx + \int_1^2 -(x^2 - 5x + 4) \\
 &= \left[\frac{x^3}{3} - \frac{5x^2}{2} + 4x \right]_0^1 + \left[-\frac{x^3}{3} + \frac{5x^2}{2} - 4x \right]_1^2 \\
 &= \left(\frac{1}{3} - \frac{5}{2} + 4 \right) + \left(-\frac{8}{3} + 10 - 8 + \frac{1}{3} - \frac{5}{2} + 4 \right) \\
 &= \left(\frac{3}{2} + \frac{1}{3} \right) + \left(6 - \frac{7}{3} - \frac{5}{2} \right) \\
 &= \frac{9+2}{6} + \left(\frac{36-14-15}{6} \right) \\
 &= \frac{11}{6} + \frac{7}{6} = \frac{18}{6} = 3
 \end{aligned}$$

Question4

The area (in sq. units) of the region bounded by the lines $x = 0, x = \frac{\pi}{2}$ and $f(x) = \sin x, g(x) = \cos x$ is

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

$2(\sqrt{2} - 1)$

B.

$2(\sqrt{2} + 1)$

C.

$2(\sqrt{3} - 1)$

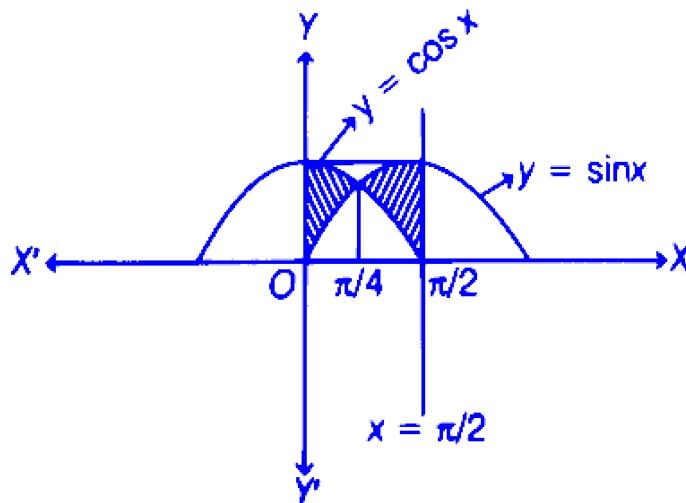
D.

$3\sqrt{2} + 1$

Answer: A**Solution:**

Required area (A)

$$= \int_0^{\frac{\pi}{4}} (\cos x - \sin x) dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin x - \cos x) dx$$



$$\begin{aligned}
 &= (\sin x + \cos x) \Big|_0^{\frac{\pi}{4}} + (-\cos x - \sin x) \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\
 &= \left[\left(\sin \frac{\pi}{4} + \cos \frac{\pi}{4} \right) - (\sin 0 + \cos 0) \right] - \left[\left(\cos \frac{\pi}{2} + \sin \frac{\pi}{2} \right) - \left(\cos \frac{\pi}{4} + \sin \frac{\pi}{4} \right) \right] \\
 &= (\sqrt{2} - 1) - (1 - \sqrt{2}) \\
 &= \sqrt{2} - 1 - 1 + \sqrt{2} \\
 &= 2\sqrt{2} - 2 = 2(\sqrt{2} - 1) \text{ sq. unit}
 \end{aligned}$$

Question5

The area of the region lying between the curves $y = \sqrt{4 - x^2}$, $y^2 = 3x$ and the Y -axis is

AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

$$\frac{\pi}{3} - \frac{1}{2\sqrt{3}}$$

B.

$$\frac{\pi}{6} + \frac{1}{2\sqrt{3}}$$

C.

$$\frac{\pi}{3} + \frac{1}{2\sqrt{3}}$$

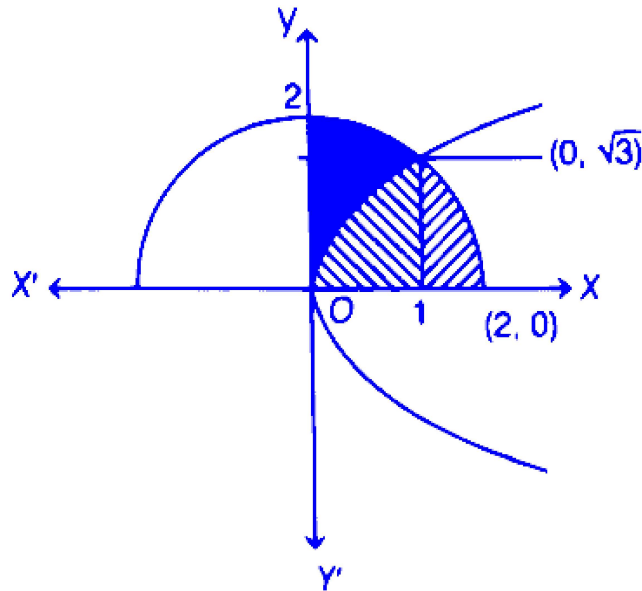
D.

$$\frac{\pi}{6} - \frac{1}{2\sqrt{3}}$$

Answer: A

Solution:

$$y = \sqrt{4 - x^2} \text{ and } y^2 = 3x$$



$$\Rightarrow 3x = 4 - x^2$$

$$\Rightarrow x^2 + 3x - 4 = 0$$

$$\Rightarrow (x + 4)(x - 1) = 0$$

$$x = 1, -4 \Rightarrow y = \sqrt{3}$$

$\therefore$ Required area,

$$= \frac{1}{3} \int_0^{\sqrt{3}} y^2 dy + \int_{\sqrt{3}}^2 \sqrt{4-y^2} dy$$

$$= \frac{1}{3} \left[\frac{y^3}{3} \right]_0^{\sqrt{3}} + \left[\frac{y}{2} \sqrt{4-y^2} + \frac{4}{2} \sin^{-1} \frac{y}{2} \right]_{\sqrt{3}}^2$$

$$= \frac{1}{9} (3\sqrt{3}) + 2 \frac{\pi}{2} - \frac{\sqrt{3}}{2} - 2 \left(\frac{\pi}{3} \right)$$

$$= \frac{\sqrt{3}}{3} - \frac{\sqrt{3}}{2} + \frac{\pi}{3} = \frac{\pi}{3} - \frac{1}{2\sqrt{3}}$$

Question6

The area of the region (in sq. units) enclosed between the curves $y = |x|$, $y = [x]$ and the ordinates $x = -1$, $x = 0$, $x = 1$ is

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

2

B.

$\frac{3}{2}$

C.

3

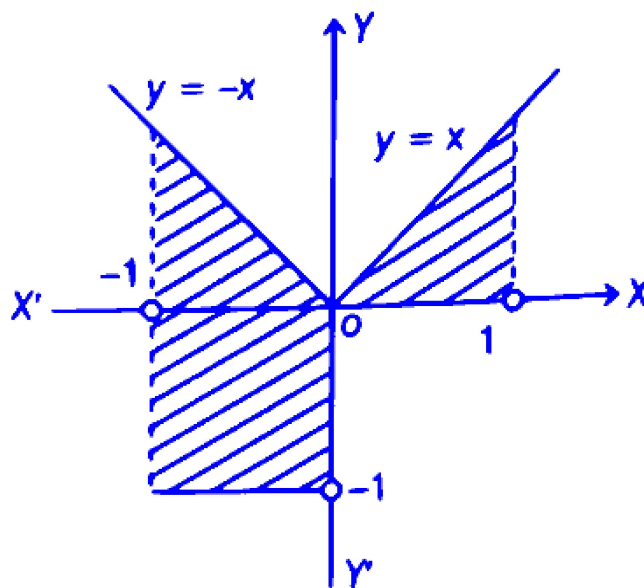
D.

$\frac{5}{2}$

Answer: A

Solution:

Area of enclosed region,



$$\begin{aligned} &= 2 \left(\frac{1}{2} \times 1 \times 1 \right) + (x) \\ &= 1 + 1 = 2 \end{aligned}$$

Question7

If (a, β) is the stationary point of the curve $y = 2x - x^2$, then the area bounded by the curves $y = 2^x$, $y = 2x - x^2$, $x = 0$ and $x = \alpha$ is

AP EAPCET 2024 - 23th May Morning Shift

Options:

A. $\frac{3 \log 2 + 4}{2}$

B. $\frac{3 + \log 4}{6}$

C. $\frac{3 - \log 4}{3 \log 2}$

D. $\frac{1}{\log 2} + \frac{3}{4}$

Answer: C

Solution:

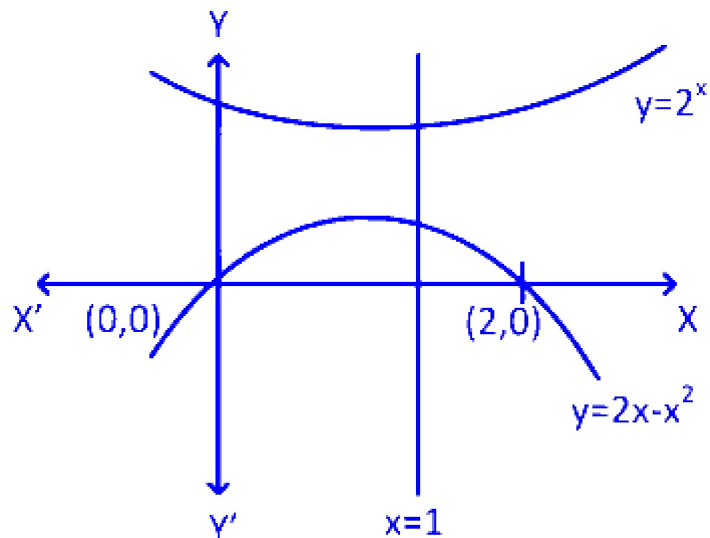
As (α, β) is the stationary point of

$$y = 2x - x^2, \frac{dy}{dx} = 0$$

$$2 - 2x$$

$$\Rightarrow x = 0$$

$$\Rightarrow \alpha = 1$$



$$\therefore \text{Required area} = \int_0^1 (y_2 - y_1) dx$$

$$= \int_0^1 (2^x - 2x + x^2) dx = \left[\frac{2^x}{\log 2} - x^2 + \frac{x^3}{3} \right]_0^1$$

$$= \left[\frac{2^1}{\log 2} - 1 + \frac{1}{3} - \frac{1}{\log 2} \right]$$

$$\begin{aligned}
 &= \frac{2}{\log 2} - 1 + \frac{1}{3} - \frac{1}{\log 2} \\
 &= \frac{1}{\log 2} - \frac{2}{3} \\
 &\Rightarrow \frac{3 - \log 4}{3 \log 2}
 \end{aligned}$$

Question8

The area (in sq units) bounded by the curves $x = y^2$ and $x = 3 - 2y^2$ is

AP EAPCET 2024 - 22th May Morning Shift

Options:

A. 8

B. $8/3$

C. 4

D. 6

Answer: C

Solution:

$$x = y^2, x = 3 - 2y^2$$

$$y = \sqrt{x}$$

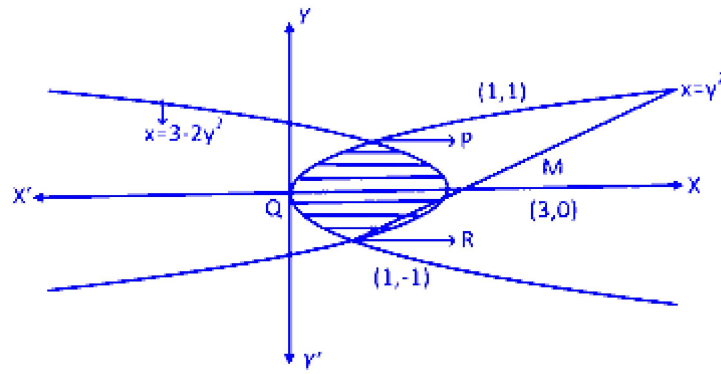
$$y = \sqrt{\frac{3-x}{2}}$$

Vertices $(0, 0)$ and $(3, 0)$

intersect at $(1, 1)$ and $(1, -1)$

Area of shaded region

$$= 2 \times (\text{Area of } PQM)$$



$$\begin{aligned}
 &= 2 \left\{ \left[\int_0^1 \sqrt{x} dx \right] + \left[\int_1^3 \sqrt{\frac{3-x}{2}} dx \right] \right\} \\
 &= 2 \left[\left(\frac{2}{3} x^{3/2} \right)_0^1 + \left(\frac{-1}{\sqrt{2}} \cdot \frac{2}{3} (3-x)^{3/2} \right)_1^3 \right] \\
 &= 2 \left[\frac{2}{3} - \left(0 - \frac{1}{\sqrt{2}} \cdot \frac{2}{3} 2^{3/2} \right) \right] \\
 &= 2 \left(\frac{2}{3} + \frac{4}{3} \right) = 2 \times \frac{6}{3} = 4
 \end{aligned}$$

Area of required region = 4sq units

Question9

Area of the region enclosed by the curves

$3x^2 - y^2 - 2xy + 4x + 1 = 0$ and $3x^2 - y^2 - 2xy + 6x + 2y = 0$ is

AP EAPCET 2024 - 21th May Evening Shift

Options:

- A. $\frac{3}{4}$
- B. $\frac{1}{4}$
- C. 1
- D. $\frac{1}{2}$

Answer: B

Solution:

Given curves are

$$3x^2 - y^2 - 2xy + 4x + 1 = 0 \quad \dots (i)$$

$$\text{and } 3x^2 - y^2 - 2xy + 6x + 2y = 0 \quad \dots (ii)$$

Eqs. (i) and (ii) can be written as

$$(3x + y + 1)(x - y + 1) = 0, \text{ and } (3x + y)(x - y + 2) = 0, \text{ respectively.}$$

Thus, the curve (i) represents the pair of lines

$$L_1 : 3x + y + 1 = 0$$

$$L_2 : x - y + 1 = 0$$

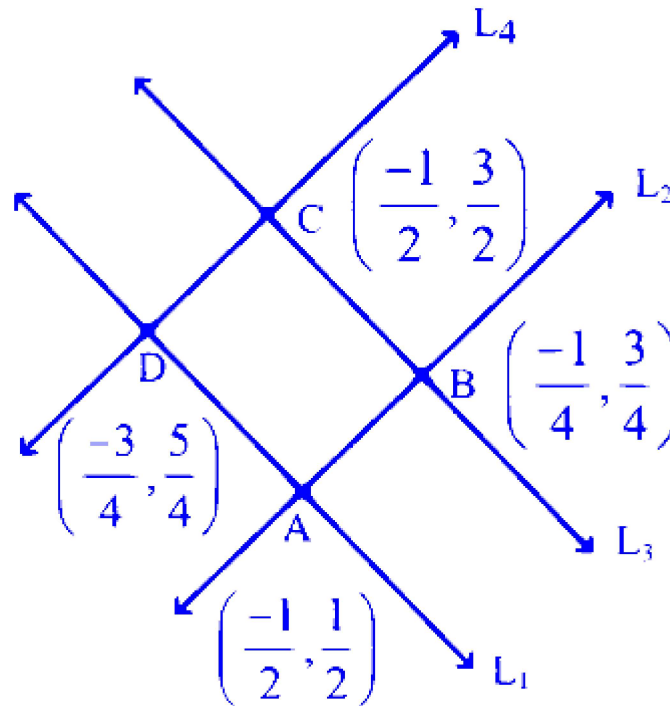
and the curve (ii) represents the pair of lines

$$L_3 : 3x + y = 0$$

$$L_4 : x - y + 2 = 0$$

Here, $L_1 \parallel L_3$ and $L_2 \parallel L_4$

So, the region closed by the given curves will be a parallelogram.



Now, required area = Area of parallelogram $ABCD = 2 \times (\text{area of } \triangle ABC)$

$$\begin{aligned}
 &= 2 \times \frac{1}{2} \begin{vmatrix} -\frac{1}{2} & \frac{1}{2} & 1 \\ -\frac{1}{4} & \frac{3}{4} & 1 \\ -\frac{1}{2} & \frac{3}{2} & 1 \end{vmatrix} \\
 &= \left[-\frac{1}{2} \left(\frac{3}{4} - \frac{3}{2} \right) - \frac{1}{2} \left(-\frac{1}{4} + \frac{1}{2} \right) + 1 \left(-\frac{3}{8} + \frac{3}{8} \right) \right] \\
 &= -\frac{1}{2} \left(-\frac{3}{4} \right) - \frac{1}{2} \left(\frac{1}{4} \right) + 0 \\
 &= \frac{3}{8} - \frac{1}{8} = \frac{2}{8} = \frac{1}{4}
 \end{aligned}$$

Question10

The area of the region under the curve $y = |\sin x - \cos x|$, $0 \leq x \leq \frac{\pi}{2}$ and above X -axis, is (in sq units)

AP EAPCET 2024 - 21th May Morning Shift

Options:

A. $2\sqrt{2}$

B. $2\sqrt{2} - 1$

C. $2(\sqrt{2} - 1)$

D. $2(\sqrt{2} + 1)$

Answer: C

Solution:

To find the area under the curve $y = |\sin x - \cos x|$ from $x = 0$ to $x = \frac{\pi}{2}$, and above the X -axis, we proceed as follows:

The expression $|\sin x - \cos x|$ changes based on x . Solving $|\sin x - \cos x| = 0$:

$$\begin{aligned}\sin x - \cos x &= 0 \\ \sin x &= \cos x \\ \tan x &= 1 \\ x &= \frac{\pi}{4}\end{aligned}$$

This implies that $\sin x - \cos x$ is negative from 0 to $\frac{\pi}{4}$ and positive from $\frac{\pi}{4}$ to $\frac{\pi}{2}$.

Thus, the function can be defined piecewise as:

$$|\sin x - \cos x| = \begin{cases} \cos x - \sin x & \text{for } 0 \leq x < \frac{\pi}{4} \\ \sin x - \cos x & \text{for } \frac{\pi}{4} \leq x \leq \frac{\pi}{2} \end{cases}$$

Integrating from $x = 0$ to $x = \frac{\pi}{4}$:

$$\int_0^{\pi/4} (\cos x - \sin x) dx = \int_0^{\pi/4} \cos x dx - \int_0^{\pi/4} \sin x dx$$

Calculating each integral:

$$\int_0^{\pi/4} \cos x dx = [\sin x]_0^{\pi/4} = \sin \frac{\pi}{4} - \sin 0 = \frac{\sqrt{2}}{2}$$

$$\int_0^{\pi/4} \sin x \, dx = [-\cos x]_0^{\pi/4} = -\cos \frac{\pi}{4} + \cos 0 = -\frac{\sqrt{2}}{2} + 1$$

Thus, the area from $x = 0$ to $x = \frac{\pi}{4}$ is:

$$\int_0^{\pi/4} (\cos x - \sin x) \, dx = \frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2} + 1\right) = \sqrt{2} - 1$$

Integrating from $x = \frac{\pi}{4}$ to $x = \frac{\pi}{2}$:

$$\int_{\pi/4}^{\pi/2} (\sin x - \cos x) \, dx = \int_{\pi/4}^{\pi/2} \sin x \, dx - \int_{\pi/4}^{\pi/2} \cos x \, dx$$

Calculating each integral:

$$\int_{\pi/4}^{\pi/2} \sin x \, dx = [-\cos x]_{\pi/4}^{\pi/2} = -\cos \frac{\pi}{2} + \cos \frac{\pi}{4} = 0 + \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{2}$$

$$\int_{\pi/4}^{\pi/2} \cos x \, dx = [\sin x]_{\pi/4}^{\pi/2} = \sin \frac{\pi}{2} - \sin \frac{\pi}{4} = 1 - \frac{\sqrt{2}}{2}$$

Thus, the area from $x = \frac{\pi}{4}$ to $x = \frac{\pi}{2}$ is:

$$\int_{\pi/4}^{\pi/2} (\sin x - \cos x) \, dx = \frac{\sqrt{2}}{2} - \left(1 - \frac{\sqrt{2}}{2}\right) = \sqrt{2} - 1$$

Total Area:

Adding the two parts:

$$(\sqrt{2} - 1) + (\sqrt{2} - 1) = 2(\sqrt{2} - 1)$$

Thus, the total area under the curve is $2(\sqrt{2} - 1)$ square units.

Question11

Area of the region (in sq units) enclosed by the curves $y^2 = 8(x + 2)$, $y^2 = 4(1 - x)$ and the Y -axis is

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. $\frac{8}{3}(5 - 3\sqrt{2})$

B. $\frac{8}{3}(\sqrt{2} - 1)$

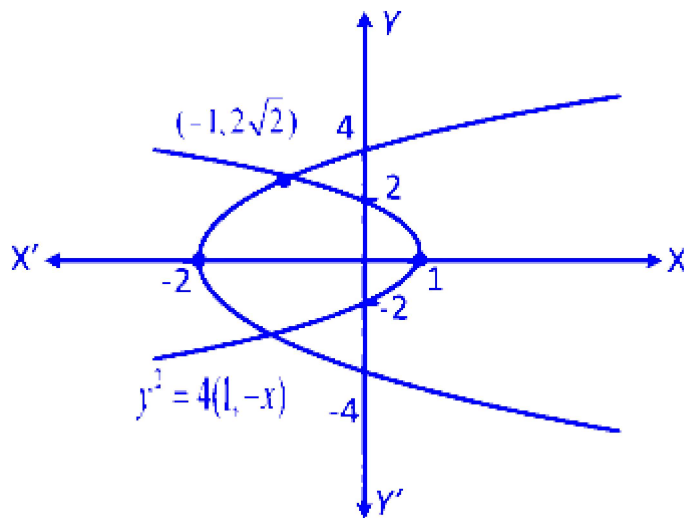
C. $\frac{8}{3}(3 - \sqrt{2})$

D. $\frac{4}{3}(\sqrt{2} + 1)$

Answer: A

Solution:

We have,



$$y^2 = 8(x + 2) \text{ and } y^2 = 4(1 - x)$$

The intersecting point of the above 2 parabola are $(-1, \pm 2\sqrt{2})$

$\therefore$ Required area

$$\begin{aligned} &= \left| 2 \left[\int_2^{2\sqrt{2}} \left(1 - \frac{y^2}{4} \right) dy + \int_{2\sqrt{2}}^4 \left(\frac{y^2}{8} - 2 \right) dy \right] \right| \\ &= \left| 2 \left[\left[y - \frac{y^3}{12} \right]_2^{2\sqrt{2}} + \left[\frac{y^3}{24} - 2y \right]_{2\sqrt{2}}^4 \right] \right| \\ &= \left| 2 \left[\left(2\sqrt{2} - \frac{16\sqrt{2}}{12} \right) - \left(2 - \frac{8}{12} \right) + \left(\frac{64}{24} - 8 \right) \right. \right. \\ &\quad \left. \left. - \left(\frac{16\sqrt{2}}{24} - 4\sqrt{2} \right) \right] \right| \\ &= \left| 2 \left[\frac{2\sqrt{2}}{3} - \frac{4}{3} - \frac{16}{3} + \frac{10\sqrt{2}}{3} \right] \right| \\ &= \left| 2 \left[\frac{12\sqrt{2}}{3} - \frac{20}{3} \right] \right| \\ &= \left| \frac{8}{3} [3\sqrt{2} - 5] \right| = \frac{8}{3} (5 - 3\sqrt{2}) \end{aligned}$$

Question12

The area (in sq units) of the smaller region lying above the X -axis and bounded between the circle $x^2 + y^2 = 2ax$ and the parabola $y^2 = ax$ is

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Options:

A. $2a^2 \left(\frac{\pi}{4} - \frac{2}{3} \right)$

B. $a^2 \left(\frac{\pi}{4} - \frac{2}{3} \right)$

C. $a^2 \left(\frac{\pi}{4} + \frac{2}{3} \right)$

D. $a^2 \left(\frac{\pi^2}{4} - \frac{1}{3} \right)$

Answer: B

Solution:

Given the problem of finding the area of a region bounded by a circle and a parabola, let's explore how to calculate it.

The equation of the circle is:

$$x^2 + y^2 = 2ax$$

This can be rewritten by completing the square for x :

$$(x - a)^2 + y^2 = a^2$$

The equation of the parabola is:

$$y^2 = ax$$

Finding Points of Intersection:

To find the intersection points between the circle and the parabola, we set:

$$x^2 + ax = 2ax$$

$$x^2 - ax = 0$$

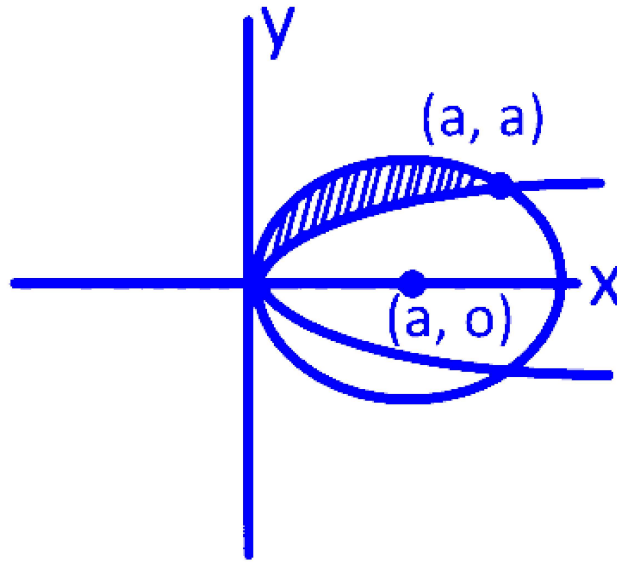
$$x(x - a) = 0$$

Thus, $x = 0$ or $x = a$.

For $x = 0$, substituting in the parabola $y^2 = ax$, we get $y = 0$.

For $x = a$, substituting in the parabola $y^2 = ax$, we get $y = \pm a$.

Thus, the points of intersection are $(0, 0)$, (a, a) , and $(a, -a)$.



Calculating the Area:

We need to integrate the area between the circle and the parabola from $x = 0$ to $x = a$:

$$\text{Required Area} = \int_0^a \left(\sqrt{a^2 - (x-a)^2} - \sqrt{a}\sqrt{x} \right) dx$$

This evaluates to:

$$= \left[\frac{x-a}{2} \sqrt{a^2 - (x-a)^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x-a}{a} \right) - \sqrt{a} \times \frac{2}{3} x^{\frac{3}{2}} \right]_0^a$$

$$= \left[-\sqrt{a} \cdot a^{\frac{3}{2}} \times \frac{2}{3} - \frac{a^2}{2} \cdot \sin^{-1}(-1) \right]$$

$$= -\frac{2}{3}a^2 + \frac{a^2}{2} \times \frac{\pi}{2}$$

Therefore, the area of the smaller region lying above the X -axis is:

$$a^2 \left(\frac{\pi}{4} - \frac{2}{3} \right)$$

Question13

The area of the region (in sq units) enclosed by the curve $y = x^3 - 19x + 30$ and the X -axis, is

AP EAPCET 2024 - 18th May Morning Shift

Options:

A. $\frac{167}{2}$

B. $\frac{517}{2}$

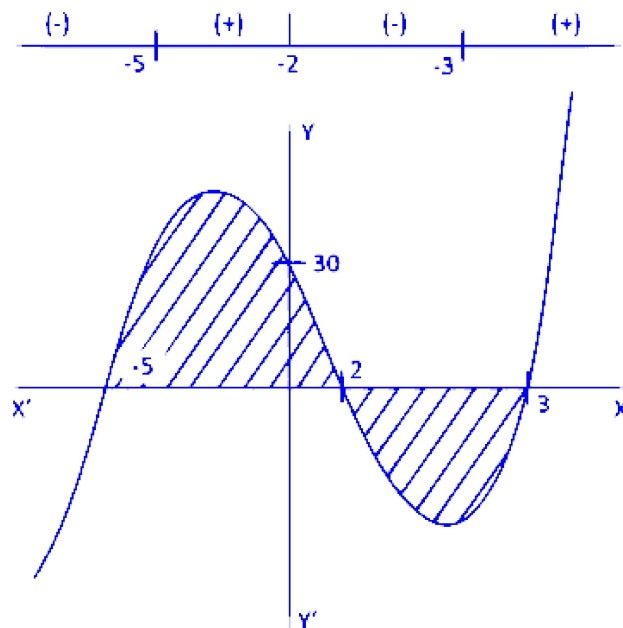
C. 36

D. 72

Answer: B

Solution:

$$\begin{aligned} (y) &= x^3 - 19x + 30 \\ &= (x - 2)(x - 3)(x + 5) \end{aligned}$$



∴ Required area

$$\begin{aligned} &= \int_{-5}^2 (x^3 - 19x + 30) dx + \left| \int_2^3 x^3 - 19x + 30 dx \right| \\ &= \left[\frac{x^4}{4} - \frac{19x^2}{2} + 30x \right]_{-5}^2 \\ &+ \left| \left[\frac{x^4}{4} - \frac{19x^2}{2} + 30x \right]_2^3 \right| \\ &= (4 - 38 + 60) - \left(\frac{625}{4} - \frac{475}{2} - 150 \right) \\ &+ \left| \left(\frac{81}{4} - \frac{171}{2} + 90 \right) - (4 - 38 + 60) \right| \\ &= 26 + \frac{925}{4} + \left| \frac{99}{4} - 26 \right| = \frac{1029}{4} + \frac{5}{4} \\ &= \frac{1034}{4} = \frac{517}{2} \end{aligned}$$
